AI Analyzer Core

Technical Portfolio Brief

Henry – Data Science Portfolio July 2026

Executive Summary

Call-center supervisors cannot review every interaction, while agents often receive late, generic feedback based on lagging KPIs. AI Analyzer Core turns dual-channel calls into auditable, turn-linked evidence and presents it through an agent KPI tracker and a supervisor review board. The portfolio release covers **100 calls, ten synthetic portfolio agents, ten calls per agent, and 11,056 fused turns**.

The system combines Faster Whisper, deterministic speaker assignment, Hugging Face sentiment, lexical service behaviors, acoustic measurements, versioned fusion, and an optional grounded local Qwen copilot. Models extract signals; deterministic contracts define meaning and own numeric truth.

Primary governance rule. Agent ordering by mean risk is descriptive, unadjusted for call mix, and **not suitable for employment decisions**. The system prioritizes evidence for human review; it does not authorize discipline, compensation, termination, or customer remediation.

Verified snapshot

Evidence	Result	Interpretation
Analytical coverage	100 calls; 11,056 turns	Exact metadata reconciliation; ten calls per agent.
Contract validation	Pass; zero blockers	Version 1.0.0 applied to canonical outputs.
Automated verification	47 tests passing	Contracts, dashboards, Copilot, outcomes, and risk pilot.
Risk distribution	76 low; 24 medium; 0 high	Uncalibrated heuristic, not proof of call quality.
Survey labels	0 real; 30 synthetic	Training remains blocked by design.
Local Copilot smoke set	6/6 grounded; 2 fallbacks	Guardrail rejected unsupported numeric claims.

Known failures visible before the design details

- `conversation_risk_v1` is hand-weighted, uncalibrated, and has not yet been compared with completed human labels.
- Estimated AHT is recording duration, not an official telephony metric; acoustic intensity is relative within each call, not an emotion label.
- Official CSAT, NPS, QA, Five Stars, and real surveys are unavailable. Demo values are explicitly synthetic fixtures.
- Agent comparisons are not adjusted for call complexity, language, market, or workload.
- Local Qwen averaged 22.49 seconds on CPU, so the current Copilot is a post-call/end-of-day prototype, not live assistance.

1 Product and System Design

1.1 Users and decisions

User	Decision	Product response
Agent	What deserves attention during this shift?	Current value, goal, trend, projection, one focus, and supporting calls.
Supervisor	Where is review useful?	Team context, evidence-based triage, positive examples, and coachable calls.
Analyst	Can the result be trusted?	Stable keys, contracts, provenance, validation, and reproducible builds.
Reviewer	Is the architecture credible under limited compute?	Pretrained models, optional LLM, deterministic calculations, and explicit boundaries.

The agent surface is intentionally a **KPI Performance Tracker**, not a bonus tracker. It supports progress toward goals without calculating compensation or encouraging optimization of one metric at the expense of service, compliance, or customer protection.

1.2 End-to-end architecture

Dual-channel audio ⟶ Faster Whisper ASR ⟶ channel-based speaker assignment

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Text sentiment and service behaviors + pitch, RMS, rate, and relative levels

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Versioned fusion ⟶ turn evidence + canonical call summaries

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Agent KPI Tracker | Supervisor Board

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Controlled retrieval ⟶ optional local Qwen ⟶ output guardrails

Governed outcomes ⟶ readiness gate ⟶ supervised modeling

Figure 1: Models produce bounded signals; contracts and guardrails control business-facing output.

1.3 Five architectural principles

1. **Models extract signals; contracts define meaning.**
2. **Deterministic calculations own numeric truth.** The LLM explains retrieved facts but does not calculate KPIs.
3. **Evidence travels with conclusions.** Reason codes resolve to call and turn identifiers.
4. **Precomputed analytics define the lightweight demo.** Dashboards need neither GPU nor model server.
5. **Unsafe absence beats fabricated certainty.** Missing real labels block outcome modeling.

1.4 Roadmap boundary

Phases 0–3 are complete: pipeline stabilization, analytical contract, agent tracker, and grounded local Copilot. The supervisor MVP and survey-readiness layer are delivered. Comparative LLM evaluation, longitudinal personalization, context-matched cross-agent learning, and calibrated outcome prediction remain explicit extensions. The authoritative phase gates live in `ROADMAP.md`.

2 Data, Models, and Lineage

2.1 Dataset and license

The sample is derived from AppTek's role-played call-center dialogue benchmark, not real customer calls. The official dataset card lists CC BY-SA 4.0 and positions the release for evaluation and analysis rather than model training or real-world customer-data applications. Portfolio publication must preserve attribution and source IDs. As a project policy, the repository does not redistribute original audio. See the official dataset card and CC BY-SA 4.0 terms.

2.2 Canonical population and data layers

`data/metadata/calls_metadata.csv` defines the 100-call population, synthetic agent assignment, channel paths, and source IDs. Every downstream coverage check must match its call keys. Per-call summaries are canonical; overlapping range summaries are excluded because they previously duplicated Agent 010.

Layer	Rows/files	Role
Metadata	100 rows	Population, source, channel, and agent registry.
Transcripts	100 files	ASR segments with timestamps and channel.
Speaker turns	100 files	Normalized agent/customer turn sequence.
Text analysis	100 files	Sentiment and lexical service behaviors.
Acoustic analysis	100 files	Pitch, RMS, speaking rate, duration, relative levels.
Fused evidence	11,056 turns	Contracted signals, reasons, and evidence IDs.
Call summary	100 rows	One canonical record per call.
Dashboards	2 JSON + 2 HTML	Agent and supervisor decision surfaces.

2.3 Model and rule choices

Stage	Implementation	Product rationale and limit
ASR	Faster Whisper `small`, CPU int8	Practical pretrained transcription; ASR errors propagate downstream.
Speaker assignment	Channel 1 agent; channel 2 customer	Deterministic for this channel-separated release, not general diarization.
Sentiment	Hugging Face RoBERTa	Text signal only; domain shift and ASR errors remain.
Service behaviors	Phrase rules for empathy, probing, ownership, frustration, next steps	Interpretable but can miss paraphrases or implicit behavior.
Acoustics	Librosa pitch, RMS, rate, duration; within-call terciles	Captures delivery separately from text; relative intensity is not emotion.
Fusion	Versioned deterministic rubric	Auditable and cheap; weights still require human validation.

Sentiment model ID: `cardiffnlp/twitter-roberta-base-sentiment-latest`.

The lightweight portfolio build begins from checked-in analytical outputs. Full raw inference remains available as a separate, heavier environment with approximately 3.9 GB of source audio and downloadable model weights.

3 Analytical Contract and Risk Rubric

3.1 Evidence classes

Class	Examples	Permitted interpretation
Observed/model signal	sentiment, pitch, RMS, rate, phrase flag	Bounded analytical signal with model or rule limitations.
Derived **PROXY**	risk, heuristic quality, estimated AHT	Review/coaching aid; never an official outcome or probability.
SYNTHETIC fixture	demo CSAT, QA, NPS, Five Stars, generated surveys	Interface, contract, join, and guardrail testing only.
REAL OUTCOME	governed business KPI or survey	Eligible for reporting/modeling only after governance checks.

The machine-readable contract at `config/analytical_contract_v1.json` fixes entity grains, primary keys, model IDs, required fields, metric semantics, reason codes, evidence references, rubric version, and declared limitations. Contract version 1.0.0 is attached to every canonical call and fused turn.

3.2 Conversation-risk heuristic

`conversation_risk_v1` combines call-level presence and absence rules, then clips the score to 0–100. Selected weights are shown because interpretability is a design requirement, not because the values are calibrated.

Risk evidence	Weight	Protective/service evidence	Weight
Customer frustration	+25	Agent empathy	-10
Negative customer sentiment	+15	Agent probing	-8
High customer intensity	+15	Agent ownership	-8
Frustration without empathy	+15	Agent next steps	-8
Missing probing/ownership/next steps	+8 each		

Bands are low (0–29), medium (30–59), and high (60–100). The observed range is 0–47, mean 19.05: 76 low, 24 medium, and zero high. This sparse upper range is a validation warning, not evidence that no difficult interactions exist.

3.3 Evidence and confidence

Stable reason codes map to turn evidence such as `call_id:turn:4`. Call summaries store each reason with its supporting turn IDs; whole-call absence rules use an empty turn list. Signal-coverage confidence measures whether inputs exist:

$$C = 0.20 I_{text} + 0.30 I_{sentiment} + \frac{0.50}{3}(I_{RMS} + I_{rate} + I_{pitch}).$$

Mean call coverage is 0.9992. A value near one means inputs were present, not that sentiment, intent, or risk was correct.

4 Agent and Supervisor Experiences

4.1 Agent KPI Performance Tracker

The offline dashboard supports all ten agents across simulated start/live/end views using the first 3, 7, and 10 calls. Each KPI exposes current value, goal, trend, projection, freshness, provenance, status, and a recommended action. It also surfaces a best analyzed call and a coachable candidate with reason and turn evidence.

Metric family	Evidence type	Boundary
Estimated AHT	Derived proxy	Recording duration; excludes holds, ACW, and telephony timing.
Risk/heuristic quality	Derived proxy	Review aid, not official quality or survey probability.
Empathy, probing, ownership, next steps	Rule evidence	Presence rate, not proof of appropriateness or effectiveness.
CSAT/QA/NPS/Five Stars	Synthetic fixture	UI demonstration only; never returned as official.

“Today’s Focus” selects one actionable area rather than optimizing every KPI at once. The interface subtly supports progress toward goals while retaining service, QA, and compliance guardrails.

4.2 Supervisor Team Performance Board

The supervisor board runs without an LLM and provides team health, shift filters, estimated-AHT-versus-risk context, behavior coverage, an explainable coaching queue, best-call examples, and coachable-call evidence. *Needs attention*, *Watch*, and *On track* order evidence review; they are not disciplinary findings or employee ratings.

4.3 Personalization and cross-agent learning boundary

Current personalization is within the available shift window: agent-specific baselines, signals, call evidence, and recommended focus. It is not yet a longitudinal learner. Context-matched cross-agent transfer remains future work because advice should compare similar call types, languages, markets, and difficulty rather than copying the highest-ranked agent. A safe future pattern is:

1. identify a recurring context and behavior with sufficient samples;
2. estimate whether the behavior is associated with better governed outcomes;
3. verify stability across agents and call-mix slices;
4. express the recommendation as a technique, not as employee imitation;
5. monitor adoption and outcome change with human review.

The complete dashboard semantics are documented in `docs/kpi_performance_tracker.md` and `docs/supervisor_board.md`.

5 Grounded Local Copilot

5.1 Trust boundary

Agent + shift + question ⟶ deterministic tool router

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KPI lookup | recap | call evidence | coaching | policy search

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Deterministic provider | local Qwen3-4B through llama.cpp

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Unsupported-number and provenance validator

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Grounded answer + citations + visible limitations

Figure 2: The LLM receives a bounded grounding packet and cannot query arbitrary project data.

The router exposes read-only tools and requires explicit agent and shift context. Retrieved policy is labeled as portfolio guidance, not employer-approved policy. The optional provider is Qwen3-4B Q4_K_M through an authenticated localhost-only llama.cpp server. The model and runtime are excluded from the lightweight package.

5.2 Numeric and provenance guardrails

Generated numeric claims are compared with the grounding packet. Unsupported values, inverted behavior coverage, or false claims of official outcomes reject the model answer and trigger deterministic wording from the same evidence. An official-CSAT question therefore withholds the synthetic value rather than disguising it as real.

5.3 Measured result and product decision

The six-prompt smoke set covered recap, AHT, the official-CSAT boundary, best call, coaching, and safe efficiency. All six final answers were grounded: four Qwen answers passed directly and two were replaced after unsupported numbers were detected.

CPU smoke set	Mean	Median	Maximum
End-to-end latency	22.49 s	20.91 s	39.43 s

This latency is unacceptable for live in-call assistance. The current Copilot is intentionally a **post-call/end-of-day explanation prototype** after deterministic artifacts already exist. A real-time product would require an explicit latency budget, cached deterministic answers, smaller or hosted inference, concurrency testing, and graceful degradation. Formal model comparison is deferred and consumes no resources unless explicitly started.

6 Outcome Readiness and Human Validation

6.1 Survey-outcome readiness

The future target is the conditional probability that a completed, linked survey is positive, not the probability that a customer responds. Training requires at least 80 completed real labels, at least 20 per class, eight agents, one instrument and label version, unique known call IDs, and `is_synthetic=False` for every eligible row.

When ready, the research baseline is standardized logistic regression with agent-grouped stratified folds and sigmoid calibration. Evaluation includes ROC AUC, average precision, Brier score, log loss, calibration bins, threshold diagnostics, and fold stability. Agent identity, raw text, survey fields, synthetic KPIs, and the heuristic risk score are excluded from features to control leakage.

The current governed source contains zero real labels. The supplied 30-row file joins to known calls but declares generated synthetic provenance, so the adapter emits `SYNTHETIC_LABELS_NOT_ALLOWED` and `NO_REAL_SURVEY_LABELS`. No model is created.

6.2 Score-blinded human review pilot

To address the lack of construct validity, the project now prepares 30 calls: the minimum, middle, and maximum risk-score call for each agent. The reviewer file hides the score, band, proxy review flag, reason codes, and weights. The separate key is joined only after at least 20 human labels are complete; 30 are preferred.

The review question is:

> *Would a supervisor benefit from reviewing this call for customer-risk context, QA context, or actionable coaching?*

The evaluator reports a confusion matrix, accuracy, balanced accuracy, precision, recall, specificity, F1, Cohen's kappa, ROC AUC, and average precision where estimable. The result must be reported even if poor.

Interpretation boundary. This is a small, purposively sampled, potentially single-reviewer pilot. It can provide exploratory construct-validity evidence, but it cannot calibrate the score, establish a production threshold, measure fairness, or authorize employment decisions. Do not tune weights and re-report the same pilot as an independent test; a revised rubric needs a new version and holdout.

The labeling protocol and commands live in `docs/risk_proxy_validation_pilot.md`.

7 Validation, Failures, and Responsible Use

7.1 Validation coverage

The lightweight package currently passes 47 unit and integration tests across seven areas: analytical contracts, duplicate-safe aggregation, agent dashboard provenance, supervisor reconciliation, Copilot grounding, survey gates, and score-blinded risk pilot preparation. The contract validator independently checks schemas, primary keys, call coverage, domains, reason evidence, versions, score types, and declared limits.

Validation target	Result	Remaining blocker
Population and lineage	Pass	Production freshness/monitoring not implemented.
Analytical contract	Pass with one warning	Risk has no completed ground truth.
Agent/supervisor dashboards	Structural and deterministic QA pass	Final browser screenshots pending.
Local Copilot smoke set	6/6 grounded	Formal comparative evaluation deferred.
Survey modeling	Correctly blocked	No governed real outcome labels.
Human risk pilot	Prepared and tested	Human labels not yet completed.

7.2 Known failure register

- Risk thresholds and weights may be disconnected from supervisor judgment; the current distribution has zero high-risk calls.
- Text sentiment and lexical behaviors inherit ASR errors and domain shift; rules can miss implicit or paraphrased behavior.
- Pitch, RMS, and speaking rate are analytical measurements, not direct labels for anger, enthusiasm, empathy, or satisfaction.
- Estimated AHT omits hold time, after-call work, transfers outside the recording, and system timing.
- Synthetic KPI fixtures demonstrate interfaces but cannot validate business impact or employee performance.
- Comparisons are unadjusted for call mix and cannot support causal statements.
- Longitudinal personalization, context-matched cross-agent learning, browser screenshots, and the final demonstration video remain incomplete.

7.3 Human, privacy, and fairness boundaries

Humans must review cited call context, apply approved policy, and record overrides. Production deployment requires legal basis and notice, minimization, redaction, retention/deletion rules, role-based access, encryption, audit logs, tenant isolation, and incident procedures. Fairness validation requires governed slices, adequate sample sizes, call-mix adjustment, subgroup error analysis, and an appeal mechanism. The portfolio does not infer protected demographics.

8 Cost, Reproducibility, and Deployment Path

8.1 Product and compute decisions

Tier	Purpose	Resource/cost boundary
Portfolio demo	Rebuild surfaces and run validation	Python 3.12 CPU; no GPU, API, database, model download, or LLM.
Full inference	Reprocess audio through ASR, text, acoustics, and fusion	Heavy dependencies, 3.9 GB audio, downloaded weights, long CPU runtime.
Optional Copilot	Explain grounded artifacts	Approximately 2.5 GB GGUF, llama.cpp, localhost server, explicit start; unsuitable live latency.
Production path	Multi-user operational service	Requires measured infrastructure, security, monitoring, concurrency, SLAs, and governed data feeds.

Precomputation is a product decision: end-of-day analysis and supervisor review can deliver portfolio value without paying the latency and infrastructure cost of live inference. The LLM is optional because the deterministic dashboards and evidence layer are the product core.

8.2 Reproduce the lightweight portfolio

```
.\scripts\setup_demo_environment.ps1
.\scripts\reproduce_portfolio.ps1
```

The locked environment runs all tests, validates the contract, rebuilds both dashboard artifacts, normalizes the synthetic survey fixture, confirms that real-label training remains blocked, verifies risk-pilot blinding and coverage, and records artifact SHA-256 hashes. Outputs are `reports/reproducibility_report.json` and `reports/reproducibility_report.md`. Semantic outputs are reproducible; timestamps, absolute environments, and optional inference runtimes may differ.

8.3 Scaling path

The same evidence contract can support encrypted object storage, queued inference workers, call/turn analytical storage, contract validation at ingestion, role-specific APIs, cached deterministic answers, optional bounded LLM explanation, and monitoring for drift, missing signals, latency, calibration, and human overrides. No deployment stage should bypass the real-outcome and human-decision gates.

9 Portfolio Handoff

Fast reviewer path

1. Read `README.md` for the 60-second business and design story.
2. Open `dashboard/kpi_performance_tracker.html` and inspect an agent across start, live, and end views.
3. Open `dashboard/supervisor_board.html` and inspect the explainable review queue.
4. Review the risk pilot protocol and blank labeling sheet.
5. Run the lightweight reproduction package and inspect the PASS report.

Deep-reference index

This brief intentionally removes the repeated catalogs, repository map, and detailed runbooks that made the previous document 19 pages. The implementation-level detail is preserved in focused, maintainable sources:

Reference	Scope
`docs/analytical_contract.md`	Entity grains, metrics, reasons, evidence, and limitations.
`docs/kpi_performance_tracker.md`	Agent views, KPI semantics, and synthetic fixtures.
`docs/supervisor_board.md`	Team triage, coaching queue, and decision boundaries.
`docs/local_customer_service_copilot.md`	Tools, provider, guardrails, smoke set, and latency.
`docs/survey_outcome_prediction.md`	Target definition, leakage controls, readiness, and evaluation.
`docs/risk_proxy_validation_pilot.md`	Human labeling and exploratory evaluation protocol.
`docs/REPRODUCIBILITY.md`	Clean setup, commands, outputs, and resource exclusions.
`ROADMAP.md`	Phase status, deferred work, and definitions of done.

Current portfolio claim

AI Analyzer Core shows how pretrained ML, interpretable heuristics, deterministic contracts, grounded LLM assistance, and role-specific product design can form a credible business solution under limited compute. Its strongest claim is not that every score is correct. It is that the architecture makes evidence, provenance, uncertainty, cost, and failure visible enough for a human to judge what is ready and what is not.

Remaining Phase 8 gates

Complete and report the 30-call human review pilot; capture final agent and supervisor screenshots; record the short walkthrough video; add the repository URL; and perform a final external portfolio review. Until then, Phase 8 remains in progress.